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$$4^{x-1} + 2^{x-3} = 2^{2x-3} + 7$$

$$\Leftrightarrow 2^{2x-2} + 2^{x-3} = 2^{2x-3} + 7$$

$$\Leftrightarrow \frac{1}{2}2^{2x} + \frac{1}{8}2^x = \frac{1}{8}2^{2x} + 7 \text{ stel } 2^x = t$$

$$\Leftrightarrow \frac{1}{4}t^2 + \frac{1}{8}t = \frac{1}{8}t^2 + 7$$

$$\Leftrightarrow \frac{1}{8}t^2 + \frac{1}{8}t - 7 = 0$$

$$\Delta = \frac{1}{64} - 4 \cdot \frac{1}{8} \cdot -7 = \frac{225}{64}$$

$$t_{1,2} = \frac{\frac{1}{8} \pm \frac{15}{8}}{\frac{1}{4}}$$

$$t_1 = \frac{14}{2} = 7$$

$$t_2 = -\frac{16}{2} = -8$$

$$2^x = -8 \text{ kan niet}$$

$$2^x = 7 \Leftrightarrow x = \log_2 7$$

References